



10 (a) Radioactive decay is both random and spontaneous.

(i) State what is meant by random.

.....
 [1]

(ii) State what is meant by spontaneous.

.....
 [1]

(iii) State **one** piece of evidence for the random nature of decay.

.....
 [1]

(b) (i) Describe the differences between nuclear fission and nuclear fusion.

.....

 [3]

(ii) Explain, with reference to the variation of binding energy per nucleon with nucleon number, why the processes of nuclear fission and nuclear fusion both result in a release of energy.

.....

 [2]

[Total: 8]





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